THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A 1 kW heater for 2 h uses: 1

(a) 2 kWh **(b)** 1 kWh **(c)** 2000 kWh **(d)** 0.5 kWh

2. A moving car has: 1

(a) only heat **(b)** no energy
(c) kinetic energy **(d)** only potential energy

3. MA = 1

(a) load/effort **(b)** effort/load **(c)** load x effort **(d)** effort - load

4. Load 100 N, effort 25 N. MA: 1

(a) 75 **(b)** 0.25 **(c)** 4 **(d)** 125

5. A stretched spring has: 1

(a) kinetic energy **(b)** no energy **(c)** potential energy **(d)** sound

6. A see-saw is a: 1

(a) screw **(b)** first class lever
(c) ramp **(d)** pulley

7. Energy of a body falling freely changes from: 1

(a) potential to kinetic **(b)** heat to light
(c) kinetic to potential **(d)** sound to heat

8. 1 kWh equals: 1

(a) 3.6 x 10^6 J **(b)** 1000 J **(c)** 36 J **(d)** 3600 J

9. Energy cannot be: 1

(a) stored **(b)** used
(c) created or destroyed **(d)** converted

10. A body at the highest point of its path has: 1

(a) no energy **(b)** maximum PE **(c)** zero PE **(d)** maximum KE

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the work done by a force of 10 N moving a body 3 m. 2

12. Define potential energy. 2

13. Name three simple machines. 2

14. Find the PE of a 3 kg body at 4 m (g = 10). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find PE of 10 kg at 3 m (g = 10). 3

16. How many units in a 2 kW heater used for 3 hours? 3

17. Why do we use a ramp to load a truck? 3

OR

Find power: 600 J in 20 s. 3

18. What is the mechanical advantage of a load 50 N lifted by effort 10 N? 3

OR

State the unit of energy and the unit of power. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain levers of the three classes with examples. 5

OR

Explain power with the formula, unit and example. 5

20. Explain the pulley. 5

OR

Explain how a machine is useful to a person with examples. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND